ECE 6360 Parallel and Heterogeneous Computing

Homework 5: Diffusion and the Heat Equation

Show that the Euler update step can be written as:

$$T_{i,j}^{n} = (1 - 4r)\, T_{i,j}^{n} + r\left(T_{i+1,j}^{n} + T_{i-1,j}^{n} + T_{i,j+1}^{n} + T_{i,j-1}^{n} \right)$$

What is r and what value for r satisfies the CFL condition if $\Delta_x = \Delta_y = h$?

What is the largest stable Δ_t for a plate with $\alpha = 1.0 \times 10^{-4}\mathrm{m}^2/\mathrm{s}$ and a spacing of $h = 1\mathrm{mm}$?